Quantum Gates and Circuits

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# Quantum Computing

Kets (Dirac Notation) and Vectors
Ket column vectors represent the states of a Quantum system
$|0\rangle$ and $|1\rangle$
Other notation is called Bra notation which is a complex conjugate of Ket vectors
$\langle 0|$ $\langle 1|$

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \; and \; |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Superposition (qubits can exist in both 0 and 1 states simultaneously) and entanglement (state of one qubit directly affects the state of another) enable quantum computers to perform computations in ways that classical computers cannot for certain problems.

# Orthonormality

Two quantum states $\langle\psi|$ and $|\phi\rangle$ are orthogonal if: $\langle\psi|\phi\rangle = 0$
That means they have no "overlap" and can be perfectly distinguished by a measurement.
Normalization condition:

$$\langle\psi|\psi\rangle = 1$$

Orthonormality of basis states $\langle 0|$ and $|1\rangle$ form an orthonormal basis, i.e., the states are normalized and orthogonal.

$$\langle 0|0\rangle = 1$$

$$\langle 1|1\rangle = 1$$

$$\langle 0|1\rangle = 0$$

$$\langle 1|0\rangle = 0$$

# Quantum Bit or Qubit

Qubits can be in either $|0\rangle$ or $|1\rangle$ or linear superposition of both states

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \textit{ where } |\alpha|^2 + |\beta|^2 = 1$$

$\alpha$ and $\beta$ are amplitudes. Square of their sum is called Quantum probability and should be 1.
The superposition is what makes Quantum computing different from classical computing.
Ground state is represented by $|0\rangle$
Excited state is represented by $|1\rangle$
Qubits in superposed state occupies all states $|0\rangle$ and $|1\rangle$ simultaneously, but collapses into $|0\rangle$ or $|1\rangle$ when an observation is made. When we do a measurement we can only get one answer and not all answers to all states in the superposition state.

# Multi Qubit

Two bits have four possible states

$$00 \quad 01 \quad 10 \quad 11$$

The state of a two-qubit system can be represented as

$$\alpha_{00}\,|00\rangle + \alpha_{01}\,|01\rangle + \alpha_{10}\,|10\rangle + \alpha_{11}\,|11\rangle$$

$$p(|00\rangle) = |\langle 00|\alpha\rangle|^2 = |\alpha_{00}|^2$$

$$\Sigma\,|\alpha|^2 = 1$$

# Qubit Basis set

Two orthogonal x-basis sets are: (Eigenstate $\sigma_x$)

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Two orthogonal y-basis sets are: (Eigenstate $\sigma_y$)

$$|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$$

$$|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle)$$

# Orthonormality of Hadamard basis

$$\langle+|+\rangle = \frac{1}{2}(\langle 0| + \langle 1|)(|0\rangle + |1\rangle)$$

$$= \frac{1}{2}(\langle 0|0\rangle + \langle 0|1\rangle + \langle 1|0\rangle + \langle 1|1\rangle)$$

$$= \frac{1}{2}(1 + 0 + 0 + 1) = 1$$

$$\langle-|-\rangle = \frac{1}{2}(\langle 0| - \langle 1|)(|0\rangle - |1\rangle)$$

$$= \frac{1}{2}(\langle 0|0\rangle - \langle 0|1\rangle - \langle 1|0\rangle + \langle 1|1\rangle)$$

$$= \frac{1}{2}(1 - 0 - 0 + 1) = 1$$

$$\langle-|+\rangle = \frac{1}{2}(\langle 0| - \langle 1|)(|0\rangle + |1\rangle)$$

$$= \frac{1}{2}(\langle 0|0\rangle + \langle 0|1\rangle - \langle 1|0\rangle - \langle 1|1\rangle)$$

$$= \frac{1}{2}(1 - 0 - 0 - 1) = 0$$

# Measurement and outcome probabilities

When we measure, we can't see qubits in superposition state. We observe the qubit either in states$|0\rangle$ or state $|1\rangle$ The amplitudes $\alpha$ and $\beta$ contain the information about the probability of each of those outcomes:

$$prob(measure and observe |0\rangle) = |\alpha|^2$$

$$prob(measure and observe |1\rangle) = |\beta|^2$$

After measurement, the qubit itself remains in the observed state. This means we can't tell right away what some original state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ might have been.

$$|\psi\rangle = \frac{1}{2}\,|0\rangle - \frac{\sqrt{3}i}{2}\,|1\rangle$$

The probability of observing it in $|1\rangle$ is

$$-\frac{\sqrt{3}i}{2} \times \frac{\sqrt{3}i}{2} = \frac{3}{4}$$

# Measurement

Measure the Qubit state below in the bases $(|0\rangle, |1\rangle)$

$$|\psi\rangle = \frac{1}{\sqrt{3}}(|0\rangle + \sqrt{2}|1\rangle)$$

$$P(0) = |\langle 0| \frac{1}{\sqrt{3}}(|0\rangle + \sqrt{2}|1\rangle)|^2 = |\frac{1}{\sqrt{3}}\langle 00\rangle + \frac{\sqrt{2}}{\sqrt{3}}\langle 01\rangle|^2 = \frac{1}{3}$$

$$P(1) = |\langle 1| \frac{1}{\sqrt{3}}(|0\rangle + \sqrt{2}|1\rangle)|^2 = |\frac{1}{\sqrt{3}}\langle 10\rangle + \frac{\sqrt{2}}{\sqrt{3}}\langle 11\rangle|^2 = \frac{2}{3}$$

Measure the Qubit state below in the bases $(|+\rangle, |-\rangle)$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$P(+) = |\frac{1}{\sqrt{2}}(\langle 0| + \langle 1|)|\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|^2 = \frac{1}{4}|\langle 00\rangle - \langle 11\rangle|^2 = 0$$

# Measurement Contd.

Measure the Qubit state below in the bases $(|+\rangle, |-\rangle)$

$$|\psi\rangle = \frac{1}{2}(|0\rangle + \frac{\sqrt{3}}{2}|1\rangle)$$

$$P(+) = |\frac{1}{\sqrt{2}}(\langle 0| + \langle 1|)|\frac{1}{2}(|0\rangle + \frac{\sqrt{3}}{2}|1\rangle)|^2 = \frac{1}{8}|\langle 00\rangle + \sqrt{3}\langle 11\rangle|^2$$

$$P(+) = \frac{(1+\sqrt{3})^2}{8} = \frac{4+2\sqrt{3}}{8}$$

$$P(-) = \frac{4-2\sqrt{3}}{8}$$

# Operations on Qubit – Unitary Matrices

Special class of matrices that preserves the length of quantum states. Quantum logic gates are represented by unitary matrices

$$U^\dagger U = UU^\dagger = I$$

where the $^\dagger$ indicates the taking complex conjugate of all elements in the transpose of U, and I is the identity matrix.

$$\alpha |0\rangle + \beta |1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$\alpha |0\rangle + \beta |1\rangle = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

Quantum Gates must be reversible and there is no way to perfectly copy a state. This is called no-cloning theorem.

# Applying Unitary Transformation

$$|\psi\rangle = \frac{1}{2}|0\rangle - \frac{\sqrt{3}i}{2}|1\rangle$$

Apply the operation

$$U = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$$

$$U|\psi\rangle = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{2} \\ \frac{-\sqrt{3}i}{2} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} \\ \frac{-i}{2} \end{pmatrix} = \frac{\sqrt{3}}{2}|0\rangle - \frac{i}{2}|1\rangle$$

The probabilities of outcomes 0 and 1 are 3/4 and 1/4 respectively. Since they sum to 1, we know that the state is normalized.

# Quantum Gates: NOT Gate

Quantum equivalent of NOT Gate is also called Pauli X Gate:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \textit{and} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

act on a single qubit. Transforms $|0\rangle$ to $|1\rangle$ and $|1\rangle$ to $|0\rangle$

$$X|0\rangle = |1\rangle \quad \textit{and} \quad X|1\rangle = |0\rangle$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \textit{and} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

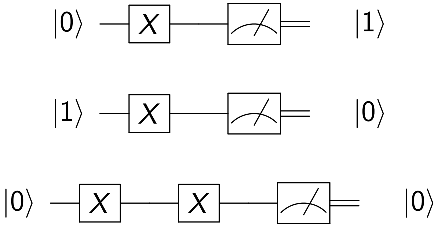
$$X(\alpha|0\rangle + \beta|1\rangle) = X \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix} = \alpha|1\rangle + \beta|0\rangle$$

| INPUT | OUTPUT |
| --- | --- |
| $|0\rangle$ | $|1\rangle$ |
| $|1\rangle$ | $|0\rangle$ |
| $\alpha|0\rangle + \beta|1\rangle$ | $\alpha|1\rangle + \beta|0\rangle$ |

# Execution: X Gate

$$X\,|+\rangle = \frac{1}{\sqrt{2}}(X\,|0\rangle + X\,|1\rangle) = \frac{1}{\sqrt{2}}(|1\rangle + |0\rangle) = |+\rangle$$

$$X\,|-\rangle = \frac{1}{\sqrt{2}}(X\,|0\rangle - X\,|1\rangle) = \frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) = -\,|-\rangle$$



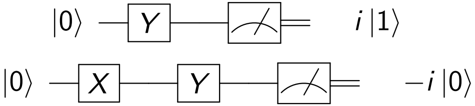
# Quantum Gates: Pauli Y Gate

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad \mathit{and} \quad \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

act on a single qubit. Transforms $|0\rangle$ to i$|1\rangle$ and $|1\rangle$ to -i$|0\rangle$

$$Y\,|0\rangle = i\,|1\rangle \quad \mathit{and} \quad Y\,|1\rangle = -i\,|0\rangle$$

$$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} \quad \mathit{and} \quad \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix}$$



# Quantum Gates: Pauli Z Gate

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \textit{and} \quad \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Mathematically when act on a single qubit, the Pauli-Z gate leaves the $|0\rangle$ alone, but flips the sign of $|1\rangle$ to -$|1\rangle$

$$Z|0\rangle = |0\rangle \quad \textit{and} \quad Z|1\rangle = -|1\rangle$$

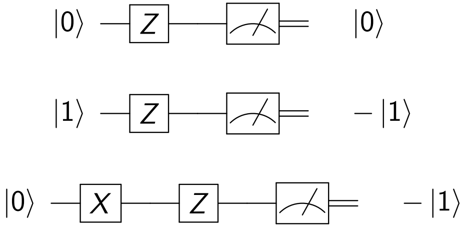
$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \textit{and} \quad \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$Z(\alpha|0\rangle + \beta|1\rangle) = \alpha Z|0\rangle + \beta Z|1\rangle = \alpha|0\rangle - \beta|1\rangle$$

| INPUT | OUTPUT |
|---|---|
| $|0\rangle$ | $|0\rangle$ |
| $|1\rangle$ | -$|1\rangle$ |
| $\alpha|0\rangle + \beta|1\rangle$ | $\alpha|1\rangle - \beta|0\rangle$ |

# Execution: Z Gate

$$Z\,|+\rangle = |-\rangle \quad \textit{and} \quad Z\,|-\rangle = |+\rangle$$



# Z Rotations

The RZ gate, or "Z rotation" is a quantum gate that modifies the relative phase between and in a more general way than the Pauli gate does. Its action depends on a parameter $\omega$ (an angle in radians). Given some initial state

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

a rotation by the amount $\omega$ is

$$RZ(\omega)\,|\psi\rangle = \alpha\,|0\rangle + \beta e^{i\omega}\,|1\rangle$$

A standard PauliZ gate maps $\alpha|0\rangle + \beta|1\rangle$ to $\alpha|0\rangle - \beta|1\rangle$. For these two operations to be identical, we need $e^{i\omega} = -1$. Using Euler's formula, we know that $e^{i\pi} = -1$

The standard matrix representation of the $RZ(\omega)$ gate the rotation around the $Z$-axis by an angle $\omega$ (in radians) is given by:

$$RZ(\omega) = \begin{pmatrix} e^{-i\omega/2} & 0 \\ 0 & e^{i\omega/2} \end{pmatrix}$$

# Factoring out the Global Phase

You will often see this matrix written in an alternative form in textbooks. Because a global phase factor does not affect quantum measurements, we can multiply the entire matrix by $e^{i\omega/2}$ to make the top-left element a real number:

$$RZ(\omega) \equiv \begin{pmatrix} 1 & 0 \\ 0 & e^{i\omega} \end{pmatrix}$$

This distinction is crucial when building multi-qubit systems or using controlled operations (like a Controlled-$RZ$). In controlled gates, that extra phase factor is only applied to specific states, transforming it from an unmeasurable global phase into a measurable relative phase!

# S and T

1. Matrix Representation of the $S$ Gate

To find the matrix representation of the $S$ gate, we substitute $\omega = \frac{\pi}{2}$ into the standard $RZ(\omega)$ matrix:

$$RZ\left(\frac{\pi}{2}\right) = \begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

To factor out the global phase, we multiply the entire matrix by $e^{i\pi/4}$:

$$S = e^{i\pi/4} \begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

2. Eigenvalues and Eigenvectors

Because the $S$ gate is a diagonal matrix, its eigenvalues and eigenvectors can be read directly from the diagonal:

Eigenvector $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ corresponds to the eigenvalue 1.

Eigenvector $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ corresponds to the eigenvalue $i$.

# Continued

3. What happens when you apply $S$ twice?

Applying the $S$ gate twice is equivalent to squaring its matrix representation:

$$S^2 = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} 1^2 & 0 \\ 0 & i^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z$$

Notice anything familiar? This is exactly the Pauli-$Z$ gate!

The second special case you are referring to occurs when $\omega = \frac{\pi}{4}$. This is known as the $T$ gate.

1. Matrix Representation of the $T$ Gate

If we substitute $\omega = \frac{\pi}{4}$ into the standard $RZ(\omega)$ matrix, we get:

$$RZ\left(\frac{\pi}{8}\right) = \begin{pmatrix} e^{-i\pi/8} & 0 \\ 0 & e^{i\pi/8} \end{pmatrix}$$

Just like with the $S$ gate, we can factor out the global phase by multiplying the entire matrix by $e^{i\pi/8}$:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

# Continued

Why is it called the "$\pi/8$ gate"?
This nickname causes a lot of confusion for beginners! It is called the $\pi/8$ gate because of its unfactored form, where the diagonal exponents contain $\pi/8$. However, when it is written in its standard, factored form, the actual phase rotation applied to the $|1\rangle$ state is $\pi/4$ radians ($45^\circ$).
2. Eigenvalues and Eigenvectors
Because the $T$ gate is also diagonal, its eigenvectors are the computational basis states, and its eigenvalues sit on the diagonal:
Eigenvector $|0\rangle$ corresponds to the eigenvalue 1.
Eigenvector $|1\rangle$ corresponds to the eigenvalue $e^{i\pi/4}$ (which can also be written as $\frac{1+i}{\sqrt{2}}$).
3. The Relationship: $T$, $S$, and $Z$
The $T$ gate represents a $45^\circ$ ($\frac{\pi}{4}$ radian) rotation around the $Z$-axis on the Bloch sphere. Because of this, it serves as the building block for the other $Z$-rotation gates:

# Continued

Applying $T$ twice gives you the $S$ gate ($45^\circ \times 2 = 90^\circ$):

$$T^2 = S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

Applying $T$ four times gives you the $Z$ gate ($45^\circ \times 4 = 180^\circ$):

$$T^4 = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

For this reason, the $T$ gate is also frequently referred to as the $\sqrt{S}$ gate or the $\sqrt[4]{Z}$ gate. It is incredibly important in fault-tolerant quantum computing because, when combined with the Hadamard gate, it allows us to approximate any single-qubit rotation with high precision!

# Continued

To implement the adjoint of the $S$ and $T$ gates using only the gates themselves, we can take advantage of the fact that these gates represent rotations around the $Z$-axis that eventually loop back to the starting point after a full $360^\circ$ ($2\pi$ radians) rotation.

1. Implementing $S^\dagger$ using only $S$ gates

The $S$ gate is a $90^\circ$ ($\pi/2$) rotation. The adjoint $S^\dagger$ is a $-90^\circ$ rotation, which is geometrically identical to rotating forward by $270^\circ$.Since $90^\circ \times 3 = 270^\circ$, you can achieve an $S^\dagger$ operation by applying the $S$ gate three times:

$$S^3 = S^\dagger$$

2. Implementing $T^\dagger$ using only $T$ gates

The $T$ gate is a $45^\circ$ ($\pi/4$) rotation. Its adjoint $T^\dagger$ is a $-45^\circ$ rotation, which is geometrically identical to rotating forward by $315^\circ$.Since $45^\circ \times 7 = 315^\circ$, you can achieve a $T^\dagger$ operation by applying the $T$ gate seven times:

$$T^7 = T^\dagger$$

# RX Gate

The matrix representation of an X rotation is

$$RX(\theta) = \begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$

$$\xrightarrow{\theta=\pi} \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = -i \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -iX$$

Rotating a qubit by $\pi$ radians ($180^\circ$) around the $X$-axis of the Bloch sphere behaves exactly like a PauliX (NOT) gate up to a global phase!

For state=0: [0.-1.j 0.+0.j] which equals $-i \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -i|1\rangle$

For state=1: [0.+0.j 0.-1.j] which equals $-i \begin{pmatrix} 1 \\ 0 \end{pmatrix} = -i|0\rangle$

# RY Gate

The matrix representation of the $RY(\theta)$ gate is unique among the primary Pauli rotations because it contains only real numbers.
The standard matrix representation of a rotation around the $Y$-axis by an angle $\theta$ is:

$$RY(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$

Why this is incredibly useful
Because the $RY$ matrix is completely real, applying it to a quantum state that has real amplitudes will keep all the amplitudes real.
For example, if you start with a qubit in the $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ state and apply $RY(\theta)$, the resulting state vector is:

$$RY(\theta)|0\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix}$$

# RY Gate contd.

This makes the $RY$ gate the go-to tool in quantum computing algorithms when you want to explore superpositions or parameterize a state on the Bloch sphere without introducing complex phases. It allows you to sweep entirely through the real circle of the Bloch sphere ($X$-$Z$ plane).
Typically, the Pauli-$Y$ gate is written as the following matrix:

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Connecting $RY(\pi)$ to the Pauli-$Y$ GateIf we substitute $\theta = \pi$ into the standard $RY(\theta)$ rotation matrix, we get:

$$RY(\pi) = \begin{pmatrix} \cos(\pi/2) & -\sin(\pi/2) \\ \sin(\pi/2) & \cos(\pi/2) \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Notice that this result is entirely real, while the standard Pauli-$Y$ matrix has imaginary components. Just like the $RZ$ and $RX$ gates we looked at earlier, this difference is entirely due to a global phase factor.

# RY Gate contd.

If we multiply the standard Pauli-$Y$ matrix by the global phase $i$, we get:

$$iY = i\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -RY(\pi)$$

Because multiplying a state vector by a global phase (like $-1$ or $i$) doesn't change any measurement probabilities, rotating by $\pi$ radians around the $Y$-axis accomplishes the exact same physical transformation as a Pauli-$Y$ gate: it performs both a bit-flip ($X$) and a phase-flip ($Z$) at the same time!

# RY Gate contd.

To prove the relationship $Y = iXZ = iRY(\pi)$, we can evaluate the matrix multiplication step-by-step using the standard definitions of the Pauli matrices and the $RY$ rotation matrix.

Part 1: Proving $Y = iXZ$

Let's write down the standard definitions for the Pauli-$X$ and Pauli-$Z$ matrices:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

First, we multiply $X$ and $Z$ together:

$$XZ = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} (0 \cdot 1 + 1 \cdot 0) & (0 \cdot 0 + 1 \cdot (-1)) \\ (1 \cdot 1 + 0 \cdot 0) & (1 \cdot 0 + 0 \cdot (-1)) \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

# RY Gate contd.

Now, we multiply the resulting matrix by the imaginary unit $i$:

$$iXZ = i\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

This matches the standard definition of the Pauli-$Y$ matrix exactly:

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Thus, $Y = iXZ$ is proven.
Part 2: Proving $Y = iRY(\pi)$The general matrix representation for an $RY(\theta)$ rotation is:

$$RY(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$

Substituting $\theta = \pi$ into the equation gives us:

$$RY(\pi) = \begin{pmatrix} \cos(\pi/2) & -\sin(\pi/2) \\ \sin(\pi/2) & \cos(\pi/2) \end{pmatrix}$$

# RY Gate contd.

Since $\cos(\pi/2) = 0$ and $\sin(\pi/2) = 1$, the matrix simplifies to:

$$RY(\pi) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Notice that this is the exact same matrix we got for $XZ$ in Part 1.
Now, if we multiply this matrix by $i$:

$$iRY(\pi) = i\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = Y$$

Thus, $Y = iRY(\pi)$ is proven.
Conclusion
By transitivity, we have successfully shown that all three expressions yield the identical matrix:

$$Y = iXZ = iRY(\pi) = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

# Quantum Gates: Hadamard Gate

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

act on a single qubit. Transform $|0\rangle$ to $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, and $|1\rangle$ to $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \ \ and \ \ H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \left[ \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right] = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \left[ \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} \right] = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

# Quantum Gates: Hadamard Gate Contd.

Hadamard gate takes a zero and turns it into an equally weighted superposition of zero and one. This state itself has a name, the plus state. When measured, there is a 50% chance of observing a zero or one. Hadamard apply to the one state creates the minor state, which is still 50% zero and one, but with the one state having a negative amplitude.

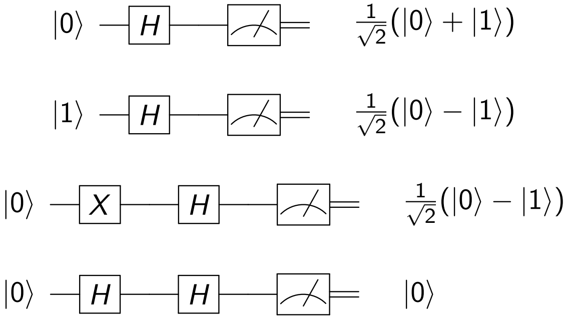
$$H(\alpha |0\rangle + \beta |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \alpha + \beta \\ \alpha - \beta \end{bmatrix}$$

| INPUT | OUTPUT |
|---|---|
| $\|0\rangle$ | $\frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$ |
| $\|1\rangle$ | $\frac{1}{\sqrt{2}} \|0\rangle - \|1\rangle$ |
| $\alpha \|0\rangle + \beta \|1\rangle$ | $\frac{\alpha}{\sqrt{2}}(\|0\rangle + \|1\rangle) + \frac{\beta}{\sqrt{2}}(\|0\rangle - \|1\rangle)$ |
| | $\frac{\alpha+\beta}{\sqrt{2}} \|0\rangle + \frac{\alpha-\beta}{\sqrt{2}} \|1\rangle$ |

# Execution: Hadamard Gate

$$HH|0\rangle = H(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}(H|0\rangle + H|1\rangle)$$

$$= \frac{1}{\sqrt{2}}(\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)+\frac{1}{\sqrt{2}}(|0\rangle-|1\rangle)) = \frac{1}{2}(|0\rangle+|1\rangle+|0\rangle-|1\rangle = |0\rangle$$



# Execution: Hadamard Gate Contd.

$$|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

$$|1\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$$

$$|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$$

# Quantum Gates: Controlled NOT Gate.

This Gate operates on 2 qubits, control qubit and target qubit. If the control qubit is 0, target is unchanged. If the control qubit is 1 then the target is inverted.

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

| INPUT | OUTPUT |
|---|---|
| $\|00\rangle$ | $\|00\rangle$ |
| $\|01\rangle$ | $\|01\rangle$ |
| $\|10\rangle$ | $\|11\rangle$ |
| $\|11\rangle$ | $\|10\rangle$ |

# Quantum Gates: Controlled NOT Gate. contd..

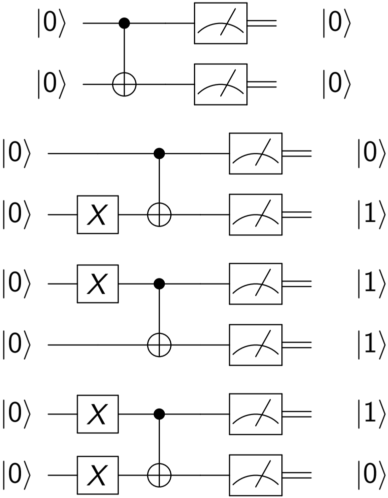
$$|00\rangle = |0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$|01\rangle = |0\rangle \otimes |1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$|10\rangle = |1\rangle \otimes |0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$|11\rangle = |1\rangle \otimes |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

# Execution: CNOT Gate



# Tensor Products

Tensor Product of $|0\rangle$ and $|1\rangle$

$$|0\rangle|1\rangle = |01\rangle = |0\rangle \otimes |1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Tensor Product of $|0\rangle$ and $|+\rangle$

$$|0\rangle|+\rangle = |0+\rangle = |0\rangle \otimes |+\rangle = |0\rangle \otimes \left( \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right) =$$
$$\frac{1}{\sqrt{2}} (|00\rangle + |01\rangle)$$

# Tensor Products Contd.

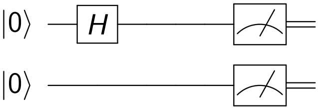
Tensor Product of $|+\rangle$ and $|1\rangle$

$|+\rangle|1\rangle = |+1\rangle = |+\rangle \otimes |1\rangle = \left(\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) \otimes |1\rangle\right) =$
$\frac{1}{\sqrt{2}}\left(|01\rangle + |11\rangle\right)$

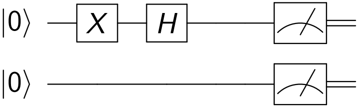
Tensor Product of $|-\rangle$ and $|+\rangle$

$|-\rangle|+\rangle = |-+\rangle = |-\rangle \otimes |+\rangle = \left(\frac{1}{\sqrt{2}}\left(|0\rangle - |1\rangle\right)\right) \otimes$
$\left(\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)\right) = \left(\frac{1}{2}\left(|00\rangle + |01\rangle - |10\rangle - |11\rangle\right)\right)$

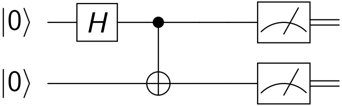
# Execution: EPR State and Entanglement



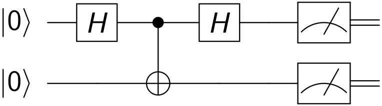

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$




$$\frac{1}{\sqrt{2}}(|00\rangle - |10\rangle)$$



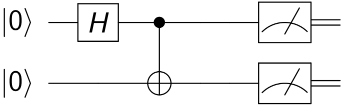

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$




$$\frac{1}{\sqrt{2}}\left[\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) + \frac{1}{\sqrt{2}}(|01\rangle - |11\rangle)\right]$$

# Entanglement

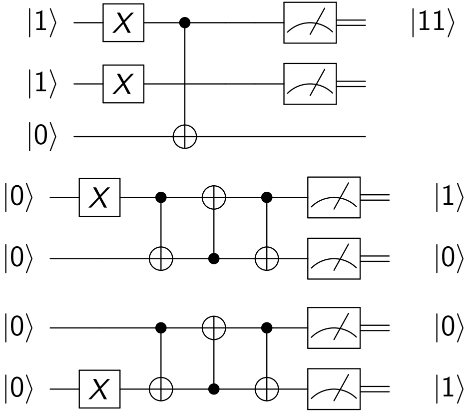
The Bell State (Maximally Entangled State)




$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

This state is an equal superposition of both qubits being in the $|0\rangle$state and both qubits being in the $|1\rangle$ state, and the important thing is that the qubits are entangled. The key feature of this state is that if you measure one qubit, you instantly know the state of the other qubit, no matter how far apart they are.

# Execution: AND Gate and SWAP Gate

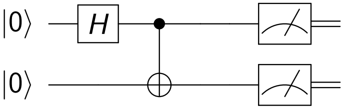


# Bell State

Tensor Product of $|+\rangle$ and $|0\rangle$

$$|+\rangle|0\rangle = |+0\rangle = |+\rangle \otimes |0\rangle = \left(\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) \otimes |0\rangle\right) =$$

$$\frac{1}{\sqrt{2}}\left(|00\rangle + |10\rangle\right) \xrightarrow{CNOT} = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) = |\phi^+\rangle$$



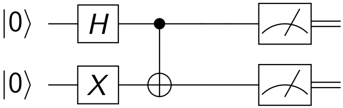

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

# Bell State Contd..

Tensor Product of $|+\rangle$ and $|1\rangle$

$$|+\rangle|1\rangle = |+1\rangle = |+\rangle \otimes |1\rangle = \left(\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) \otimes |1\rangle\right) =$$

$$\frac{1}{\sqrt{2}}\left(|01\rangle + |11\rangle\right) \xrightarrow{CNOT} = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right) = |\psi^+\rangle$$



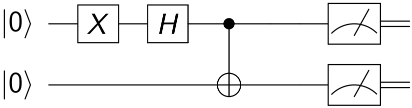

$$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

# Bell State Contd..

Tensor Product of $|-\rangle$ and $|0\rangle$

$|-\rangle|0\rangle = |-0\rangle = |-\rangle \otimes |0\rangle = \left(\frac{1}{\sqrt{2}}\left(|0\rangle - |1\rangle\right) \otimes |0\rangle\right) =$

$\frac{1}{\sqrt{2}}\left(|00\rangle - |10\rangle\right) \xrightarrow{CNOT} = \frac{1}{\sqrt{2}}\left(|00\rangle - |11\rangle\right) = |\phi^-\rangle$



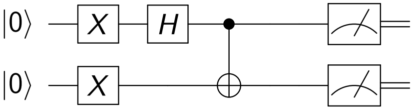

$\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$

# Bell State Contd..

Tensor Product of $|-\rangle$ and $|1\rangle$

$$|-\rangle|1\rangle = |-1\rangle = |-\rangle \otimes |1\rangle = \left(\frac{1}{\sqrt{2}}\left(|0\rangle - |1\rangle\right) \otimes |1\rangle\right) =$$

$$\frac{1}{\sqrt{2}}\left(|01\rangle - |11\rangle\right) \xrightarrow{CNOT} = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right) = |\psi^{-}\rangle$$

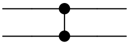



$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle\,;\ \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle\,;\ \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle\,;\ \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle$$

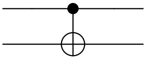
# Controlled-Z and Controlled-X

Controlled-Z



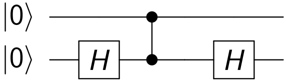
$$
\begin{aligned}
CZ(1,0)|00\rangle &= |00\rangle \\
CZ(1,0)|01\rangle &= |01\rangle \\
CZ(1,0)|10\rangle &= |10\rangle \\
CZ(1,0)|11\rangle &= -|11\rangle
\end{aligned}
$$

Controlled-X



$$
\begin{aligned}
CX(1,0)|00\rangle &= |00\rangle \\
CX1,0)|01\rangle &= |01\rangle \\
CX(1,0)|10\rangle &= |11\rangle \\
CX(1,0)|11\rangle &= |10\rangle
\end{aligned}
$$

# More Gates and Circuits




$$|00\rangle \xrightarrow{H} |0\rangle \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

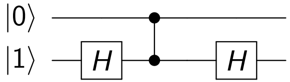
$$= \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$$

$$\xrightarrow{CZ} \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$$

$$\xrightarrow{H} \frac{1}{\sqrt{2}}\left(|0\rangle \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) + (|0\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right)$$

$$= \frac{1}{2}2(|00\rangle) = |00\rangle$$

# More Gates and Circuits




$$|00\rangle \xrightarrow{H} |0\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

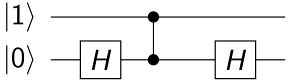
$$= \frac{1}{\sqrt{2}}(|00\rangle - |01\rangle)$$

$$\xrightarrow{CZ} \frac{1}{\sqrt{2}}(|00\rangle - |01\rangle)$$

$$\xrightarrow{H} \frac{1}{\sqrt{2}}\left(|0\rangle \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) - (|0\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right)$$

$$= \frac{1}{2}2(|01\rangle) = |01\rangle$$

# More Gates and Circuits




$$|10\rangle \xrightarrow{H} |1\rangle \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

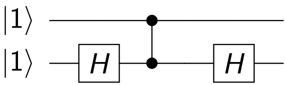
$$= \frac{1}{\sqrt{2}}(|10\rangle + |11\rangle)$$

$$\xrightarrow{CZ} \frac{1}{\sqrt{2}}(|10\rangle - |11\rangle)$$

$$\xrightarrow{H} \frac{1}{\sqrt{2}}\left(|1\rangle \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) - (|1\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right)$$

$$= \frac{1}{2}2(|11\rangle) = |11\rangle$$

# More Gates and Circuits




$$|10\rangle \xrightarrow{H} |1\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$= \frac{1}{\sqrt{2}}(|10\rangle - |11\rangle)$$

$$\xrightarrow{CZ} \frac{1}{\sqrt{2}}(|10\rangle + |11\rangle)$$

$$\xrightarrow{H} \frac{1}{\sqrt{2}}\left(|1\rangle \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) + (|1\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right)$$

$$= \frac{1}{2}2(|10\rangle) = |10\rangle$$

# Gate Operation Summary

$$X|0\rangle = |1\rangle\,; X|1\rangle = |0\rangle\,; X|+\rangle = |+\rangle\,; X|-\rangle = -|-\rangle$$

$$Y|0\rangle = i|1\rangle \quad and \quad Y|1\rangle = -i|0\rangle$$

$$Z|0\rangle = |0\rangle\,; Z|1\rangle = -|1\rangle\,; Z|+\rangle = |-\rangle\,; Z|-\rangle = |+\rangle$$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle\,; H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$

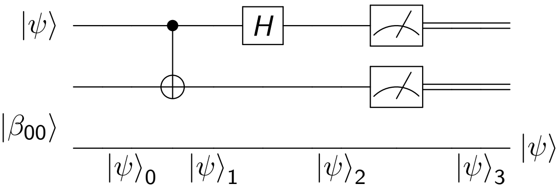
# Quantum Teleportation

The state to be transported by Alice is

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

where $\alpha$ and $\beta$ are unknown amplitudes.
The state input into the circuit $|\psi_0\rangle$ is $|\psi_0\rangle = |\psi\rangle\,|\beta_{00}\rangle$ The entangled state that is known to both Alice and Bob is

$$\frac{1}{\sqrt{2}}(|0\rangle_A\,|0\rangle_B + |1_A\rangle\,|1\rangle_B)$$



Quantum circuit for teleporting a qubit. The two top lines represent Alice's system, while the bottom line is Bob's system. The meters represent measurement, and the double lines coming out of them carry classical bits (recall that single lines denote qubits).

# Quantum Teleportation contd...

The combined state of three qubits is given by
$\alpha|0\rangle + \beta|1\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle_A|0\rangle_B + |1_A\rangle|1\rangle_B)$

$$= \frac{1}{\sqrt{2}}\left[\alpha|0\rangle(|0\rangle_A|0\rangle_B + |1_A\rangle|1\rangle_B) + \beta|1\rangle(|0\rangle_A|0\rangle_B + |1_A\rangle|1\rangle_B)\right]$$

After CNOT on the first qubit where we use the convention that the first two qubits (on the left) belong to Alice, and the third qubit to Bob. $CX(1,0)|10\rangle = |11\rangle$ and $CX(1,0)|11\rangle = |01\rangle$

$$= \frac{1}{\sqrt{2}}\left[\alpha|0\rangle(|0\rangle_A|0\rangle_B + |1_A\rangle|1\rangle_B) + \beta|1\rangle(|1\rangle_A|0\rangle_B + |0_A\rangle|1\rangle_B)\right]$$

After Hadamard gate on Alice's first qubit we get .

$$= \frac{1}{2}\left[\alpha(|0\rangle + |1\rangle)(|0\rangle_A|0\rangle_B + |1_A\rangle|1\rangle_B) + \beta(|0\rangle - |1\rangle)(|1\rangle_A|0\rangle_B + |0_A\rangle|1\rangle_B)\right]$$

# Quantum Teleportation contd...

$$= \frac{1}{2}\left[\alpha(|00\rangle_A |0\rangle_B + |01\rangle_A |1\rangle_B + |10\rangle_A |0\rangle_B + |11\rangle_A |1\rangle_B \right])$$

$$+\frac{1}{2}\left[\beta(|01\rangle_A |0\rangle_B + |00\rangle_A |1\rangle_B - |10\rangle_A |1\rangle_B) - |11\rangle_A |0\rangle_B)\right]$$

The new state is re-written by regrouping terms as follows:

$$= \frac{1}{2}\left[\, |00\rangle_A (\alpha |0\rangle_B + \beta |1\rangle_B) + |01\rangle_A (\alpha |1\rangle_B + \beta |0\rangle_B)\right.$$

$$\left. +\frac{1}{2}\left[\, |10\rangle_A (\alpha |0\rangle_B - \beta |1\rangle_B) + |11\rangle_A (\alpha |1_B\rangle - \beta |0\rangle_B)\right]\right.$$

The above expression breaks down into four terms. The first term has Alice's qubits in the state $|00\rangle$, and Bob has to do nothing If Alice performs a measurement and obtains the result 00. In that case Bob's qubit is in the state $\alpha |0\rangle + \beta |1\rangle$, which is the original state $|\psi\rangle$. If Bob receives 01 from Alice then Bob has to swap the coeffients around by applying the X gate. If the measurement is 10 then Bob can fix up his state by applying the Z gate. If the measurement is 11 then Bob can fix up his state by applying first an X andthen a Z gate. All four different measurement scenario and operations that Bob has to perform are below.

# Quantum Teleportation contd...

$$
\begin{aligned}
00 \to |\psi_3(00)\rangle &= \alpha\,|0\rangle_B + \beta\,|1\rangle_B \\
01 \to |\psi_3(01)\rangle &= \alpha\,|1\rangle_B + \beta\,|0\rangle_B \xrightarrow{X} \alpha\,|0\rangle_B + \beta\,|1\rangle_B \\
10 \to |\psi_3(10)\rangle &= \alpha\,|0\rangle_B - \beta\,|1\rangle_B \xrightarrow{Z} \alpha\,|0\rangle_B + \beta\,|1\rangle_B \\
11 \to |\psi_3(11)\rangle &= \alpha\,|1_B\rangle - \beta\,|0\rangle_B \xrightarrow{X} \alpha\,|0\rangle_B - \beta\,|1\rangle_B \xrightarrow{Z} \\
&= \alpha\,|0\rangle_B + \beta\,|1\rangle_B
\end{aligned}
$$

Depending on Alice's measurement outcome, Bob's qubit will end up in one of these four possible states. Of course, to know which state it is in, Bob must be told the result of Alice's measurement which is done through classical communication channel.
Once Bob has learned the measurement outcome, he can 'fix up' his state, recovering $|\psi\rangle$, by applying the appropriate quantum gate operations as explained above. After the teleportation process only the target qubit is left in the state $|\psi\rangle$, and the original data qubit ends up in one of the computational basis states $|0\rangle$ or $|1\rangle$ depending up on the measurement.